

Report 2 - Device Info

September 29, 2026

0.1 Introduction

The goal of this lab work is to get to know my GPU (more like Google Colab's).

0.1.1 Tools

- CUDA SDK
- Numba python library
- Google Colab

0.2 The work

0.2.1 First step

After making sure CUDA drivers and the numba library are installed, we can start checking numba functions and start working.

```
from numba import cuda as c
```

```
GPU_list = c.detect()
```

```
print(GPU_list)
```

The output being:

```
Found 1 CUDA devices
id 0          Tesla T4          [SUPPORTED]
          Compute Capability: 7.5
          PCI Device ID: 4
          PCI Bus ID: 0
          UUID: GPU-985cdd11-0877-cf3e-8001-827d502c8e5e
          Watchdog: Disabled
          FP32/FP64 Performance Ratio: 32
Summary:
    1/1 devices are supported
True
```

0.2.2 Choosing the GPU and getting the attributes

```
c.select_device(0)
GPU=c.get_current_device()
print(GPU.name)
print(GPU.MULTIPROCESSOR_COUNT)
```

Which outputs: Tesla T4 and 40

0.2.3 Memory

```
free_memory, total_memory = c.current_context().get_memory_info()
print(total_memory/1000000000, "GB")
```

The output is: 15.637086208 GB

0.2.4 Processor count

For that one I use Stackoverflow question 63823395 answer

```
cc_cores_per_SM_dict = {
    (2,0) : 32,
    (2,1) : 48,
    (3,0) : 192,
    (3,5) : 192,
    (3,7) : 192,
    (5,0) : 128,
    (5,2) : 128,
    (6,0) : 64,
    (6,1) : 128,
    (7,0) : 64,
    (7,5) : 64,
    (8,0) : 64,
    (8,6) : 128,
    (8,9) : 128,
    (9,0) : 128,
    (10,0) : 128,
    (12,0) : 128
}
my_sms = GPU.MULTIPROCESSOR_COUNT
my_cc = GPU.compute_capability
cores_per_sm = cc_cores_per_SM_dict.get(my_cc)
total_cores = cores_per_sm*my_sms
print("GPU-compute-capability:-" , my_cc)
print("GPU-total-number-of-SMs:-" , my_sms)
print("total-cores:-" , total_cores)
```

And the result is :

```
GPU compute capability: ComputeCapability(major=7, minor=5)
GPU total number of SMs: 40
total cores: 2560
```